

Suggested Teaching Guidelines for
IT Infrastructure and Management and DevOps PG-DITISS August 2025

Duration: 46 Classroom hours + 44 Lab hours (90 Hrs)

Objective: To introduce the student to Concepts of ITIL and Data center management.

Evaluation Method: Theory exam – 40% weightage
Lab exam - 40% weightage
Internal assessment – 20% weightage

List of Books / Other training material

Courseware: Cloud Computing Black Book, Kogent Learning, Kailash , (Wiley),2024,

Note: Every Session includes 2 Hours of theory and 2 Hours of Lab, unless indicated Otherwise.

A. ITIM (14 Theory Hrs + 2 Lab Hrs)

a. Introduction to ITIL (Each Session from 1 to 5 includes 2 Hrs of Theory only)

Session 1:

Service Strategy

- General strategy
- Competition and market space
- Service provider types
- Service management as a strategic asset
- Key process activities,
- Financial management
- Service portfolio management
- Demand management
- Key roles and responsibilities of staff

Session 2:

Service Design

- Design of architecture, processes, policies, documentation, and allowing for future business requirements
- Service Design Package (SDP)
- Service catalog management
- Service level management
- Designing for capacity management
- IT service continuity
- Information Security

Session 3:

Service Transition

- Service Asset and Configuration Management
- Transition Planning and Support

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- Release and deployment management
- Change Management

- Knowledge Management
- key roles of staff

Session 4:

Service Operation

- Balancing conflicting goals (e.g. reliability Vs cost etc)
- Event management
- Incident management
- Problem management
- Event fulfillment
- Asset management
- Service desk
- Technical and application management
- Key roles and responsibilities for staff

Session 5:

Continual Service Improvement (CSI)

- Training and awareness
- Ongoing scheduling
- Roles created
- Ownership assigned
- Activities identified to be successful

b. Data Center Management:

Session 6: (2 Theory Hrs)

Introduction

- Data Center Architecture, Requirements & prerequisites
- Required Physical Area for Equipment and Unoccupied Space
- Required power to run all the devices
- Required cooling and HVAC
- Required weight
- Required Network bandwidth
- Budget Constraints
- Selecting a Geographic Location
- Safe from Natural hazards & Manmade disasters
- Availability of local technical talent
- Abundant and Inexpensive Utilities Such as Power and Water
- Selecting and Existing building
- Characteristics of an Outstanding Design
- Guidelines for Planning a Data Center
- Data Centre structures

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- Raised Floor Design and Deployment
- Design and Plan against Vandalism

Session 7: Infrastructure in a data center (2 Theory Hrs + 2 Lab Hrs)

- Modular Cabling Design
- Points of Distribution
- ISP Network Infrastructure and WAN Links
- Network Operations Center and Monitoring
- Data center physical security, Logical security, and cleaning
- Reasons for data center Consolidation
- Consolidation opportunities
- Datacenter Servers
- Server Capacity Planning
- Disaster Recovery
- Data Center Security Guidelines
- Internet Security Guidelines
- Internet security
- Source Security Issues
- Best Practices for System Administration
- System Administration Work Automation

B. DevOps (32 Theory Hrs + 42 Lab Hrs)

Session 8:

- Introduction of virtualization
- Virtualization types: type1, type2
- Virtualization, Hardware Virtualization, Para-Virtualization, Cloning, Snapshot, and Template
- Operating System Virtualization
- Cluster Architecture
- Cluster Requirements

Assignment –Lab

- Create and configure VM using VBox
- Deploy code on VM

Session 9: Storage Area Network

- Configuring a SAN (FreeNAS)
- Using SAN for high availability

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- ZFS Volume Configuration
- IP-Based Storage Communication
- Object Storage services

Lab Assignment:

- Download and install FreeNAS (TrueNAS CORE) in a virtual machine using VirtualBox/VMware Workstation. Configure a storage pool and create a ZFS dataset to be used as a SAN.
- Set up two instances of FreeNAS in virtual machines using VirtualBox/VMware Workstation. Configure iSCSI targets on both instances and set up multipath I/O (MPIO) host machine to simulate high availability.
- Within the FreeNAS virtual machine, create a ZFS pool using multiple virtual disks, add datasets, and configure snapshots and replication. Use the FreeNAS web interface to manage the ZFS volumes.
- Configure NFS and CIFS/SMB shares on the FreeNAS virtual machine. Use host PC to connect to these shares, map network drives, and transfer files.
- Install MinIO, an open-source object storage server, within a virtual machine. Set up a bucket using the MinIO web interface/command-line tool, upload objects, and configure access policies.

Session 10: (2 Theory Hrs + 4 Lab Hrs)

- Introduction to Cloud
- Cloud Computing, Cloud SPI Model, Cloud Computing Deployment Model(Public, Private, and Hybrid), Cloud Security (SLA and IAM).
- Cloud architecture
- Service models: IaaS, PaaS, SaaS
- Services provided by Cloud (Compute, Database, Developer Tools, Storage, Media, Mobile, Web, Security, Integration, etc.)
- Cloud development best practices
- Introduction to Openstack
- HCI & Its Comparison to cloud
- SDN

Assignment –Lab

- Exploring various services provided by cloud providers like App Services, Web apps, API Apps, Search, Database Servers on VMs, VM Scale Sets, Bot Services, and other cloud applications.
- HCI Mandatory & Optional
- Virtual Network configuration using SDN

Session 11 & 12:

- Cloud API integration
- DC/DR Migration
- DC/DR Storage Synchronization

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- Bootstrapping Chef/puppet Server
- Migration of physical servers to clouds

Lab Assignment:

- Cloud API Integration for DC/DR Setup
- Automated Migration and Storage Synchronization
- Configuring Chef/Puppet for Cloud Migration
- Physical to Cloud Server Migration
- Implementing High Availability with Cloud API and Configuration Management

Session 13: (2 Theory Hrs + 4 Lab Hrs)

- Centralized logging
- Nagios
- Prometheus Next Gen NMS
- Identifying bottlenecks
- Auto-scaling Auto-rebuilding cloud instances
- Updating servers without downtime
- Auto-healing
- Cloud Enable Data Center Case Study

Lab Assignment:

- Configuring Nagios on Linux server and adding Linux client for monitoring.
- Configuring Nagios on Linux server and adding Windows client for monitoring.

Session 14:

- Agile
- Agile methodologies: Scrum and Kanban
- Lean
- Implementation of Lean
- Lean and Agile in DevOps

Lab Assignment:

- Agile Fundamentals
- Agile Methodologies - Scrum
- Lean Implementation
- Lean and Agile in DevOps

Session 15 & 16:

- Introduction to DevOps
- DevOps ecosystem
- DevOps phases
- CAMS model, Kaizen, Immutable deployment

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- CI/CD Pipelines
- IAM, LXC, Docker, KVM

Lab Assignment:

- Create an httpd container running on port 8000. Name the container as web5. Do not map any directory. Create an “index.html” file & copy this file inside the container and check if the website displays your webpage. Again, modify the “index.html” file and copy it inside the container, and check if the website is updated or not.
- Create a C program(exe). Test the program on-base machine. Then create an image using Dockerfile to execute this exe program. Give image name as “Name: v1”.
- Try the above with a program that requires user input.

Session 17 & 18: Github and Jenkins

- Introduction
 - Introduction to Git
 - Core Concepts
 - Going Command Line
- Basic Git workflow with Github
 - Welcome to GitHub
 - Setup the Project Folder
 - Git Configuration (User Name and Email)
 - Copy the Repository from GitHub to Your Local Computer (git clone)
 - The First Commit
 - Publishing Changes back to GitHub (push)
- Introduction of Jenkins
- CI/CD pipeline using Jenkins.

Lab Assignment:

- Create a C program to take input from the user and test it. Then create an image using Dockerfile. Test image by the running container. Then push the image to the docker hub repository. Then also export the image in a “cprog-back.tar” file.
- Create a CI/CD pipeline using Jenkins, Docker, and GitHub to automate the web by using the NGINX image.
- Create a CI/CD pipeline using Jenkins, Docker, and GitHub to automate the web by using the httpd image.

Session 19: (2 Theory Hrs + 4 Lab Hrs)

- Introduction to AWS
- Services provided by AWS: EC2, Lambda, S3
- Introduction Virtual Private Cloud (VPC) Setup

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- Create AWS EC2 instance
- Create AWS Lambda
- Create AWS S3 bucket
- Create AWS VPC
- Create a new VPC by the name ditiss-lab with public & private subnet. Assign network address as 172.20.0.0/16. Assign 172.20.5.0/24 to the public and 172.20.10.0/24 to the private subnet. Create one instance and connect to a private subnet. Create one instance and connect to the public subnet. Install httpd on the private instance and check with curl from the public connected instance.

Session 20 & 21:

- Version Control system
- Infrastructure as Code
- Containerization with Docker
- Container Orchestration: Kubernetes, Dockerswam
- Micro Service Deployment

Lab Assignments:

- Create a new docker image using NGINX. Push it to docker hub. Create a service on Dockerswarm. Keep initial replicas as 1 and then increase the replicas to 10. Reduce the replicas to 2. Create a new image by modifying “index.html”. Push to docker hub and then update the service for a new image.
- Configure a Linux as a Master node using Kubernetes and make 2 to 3 worker node clusters.

Session 22: (2 Theory Hrs + 4 Lab Hrs)

- Introduction to Ansible and Configuration Management
- Setting Up Ansible Environment
- Ansible Playbooks and YAML Basics
- Managing Ansible Inventory
- Ansible Roles and Reusability

Lab Assignments:

- Setting Up Ansible Environment
 - Install Ansible on a local machine or a virtual machine.
 - Configure SSH keys for passwordless authentication to a target node.
 - Verify the setup by running the ansible -m ping all command.
- Writing and Executing Ansible Playbooks
 - Create a simple Ansible playbook to install and start a web server (e.g., Apache or Nginx) on a target node.
 - Use the playbook to ensure the web server is running and serving a default webpage.

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- Verify the setup by accessing the webpage from a browser.
- Managing Ansible Inventory and Roles
 - Create a static inventory file listing multiple target nodes.
 - Develop a role to install and configure a database server (e.g., MySQL or PostgreSQL).
 - Apply the role using an Ansible playbook to deploy the database server on one of the target nodes listed in the inventory.

Session 23: (2 Theory Hrs + 4 Lab Hrs)

- Introduction to Infrastructure as Code (IaC) and Terraform
- Setting Up the Terraform Environment
- Writing and Organizing Terraform Configuration Files
- Terraform State Management
- Terraform Modules and Reusability

Lab Assignments:

- Setting Up the Terraform Environment
 - Install Terraform on a local machine.
 - Configure AWS credentials for Terraform to interact with AWS services.
 - Create and apply a basic Terraform configuration to provision an EC2 instance in AWS.
- Writing and Organizing Terraform Configuration Files
 - Write a Terraform configuration file to provision a VPC, subnet, and an EC2 instance within the subnet.
 - Use variables and outputs to make the configuration more dynamic and reusable.
 - Apply the configuration and verify the resources are created in AWS.
- Terraform State Management and Modules
 - Create a Terraform module for provisioning an EC2 instance with a security group.
 - Use the module in a Terraform configuration to provision multiple EC2 instances in different subnets.
 - Manage the Terraform state using a remote backend (e.g., AWS S3) to enable collaboration and state sharing.